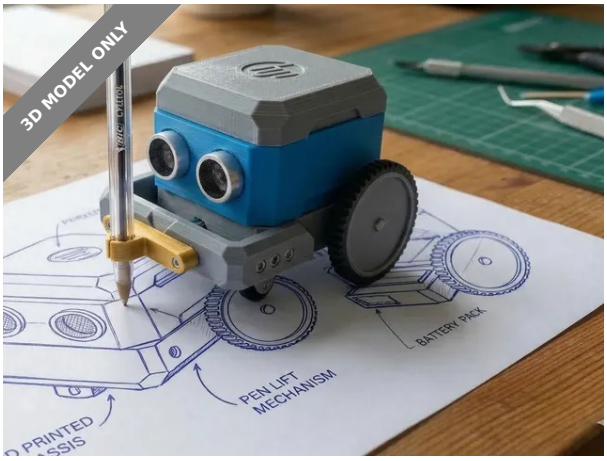


Otto's Pen Adapter

tiago

VIEW IN BROWSER

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Summary

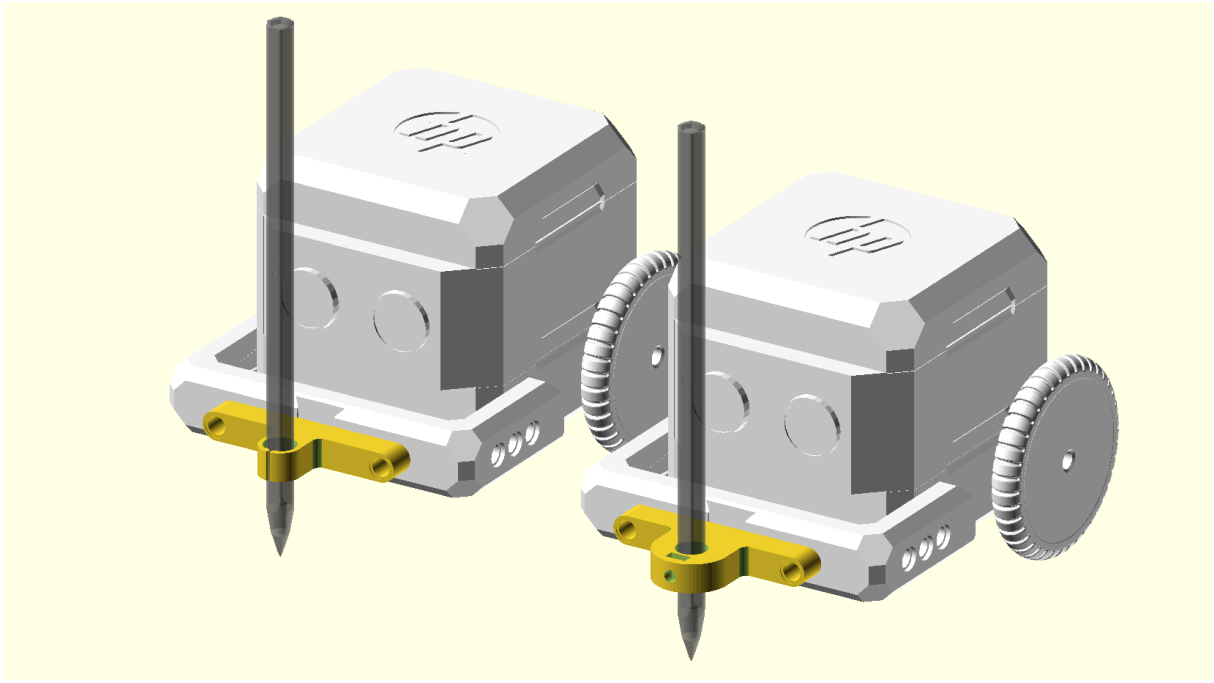
Otto's pen adapter for different pen sizes, and two locking mechanisms.

Hobby & Makers > RC & Robotics

Tags: hack pen robot educational robotics penplotter drawing ottodiy otto penadapter hpotto hpottoremix

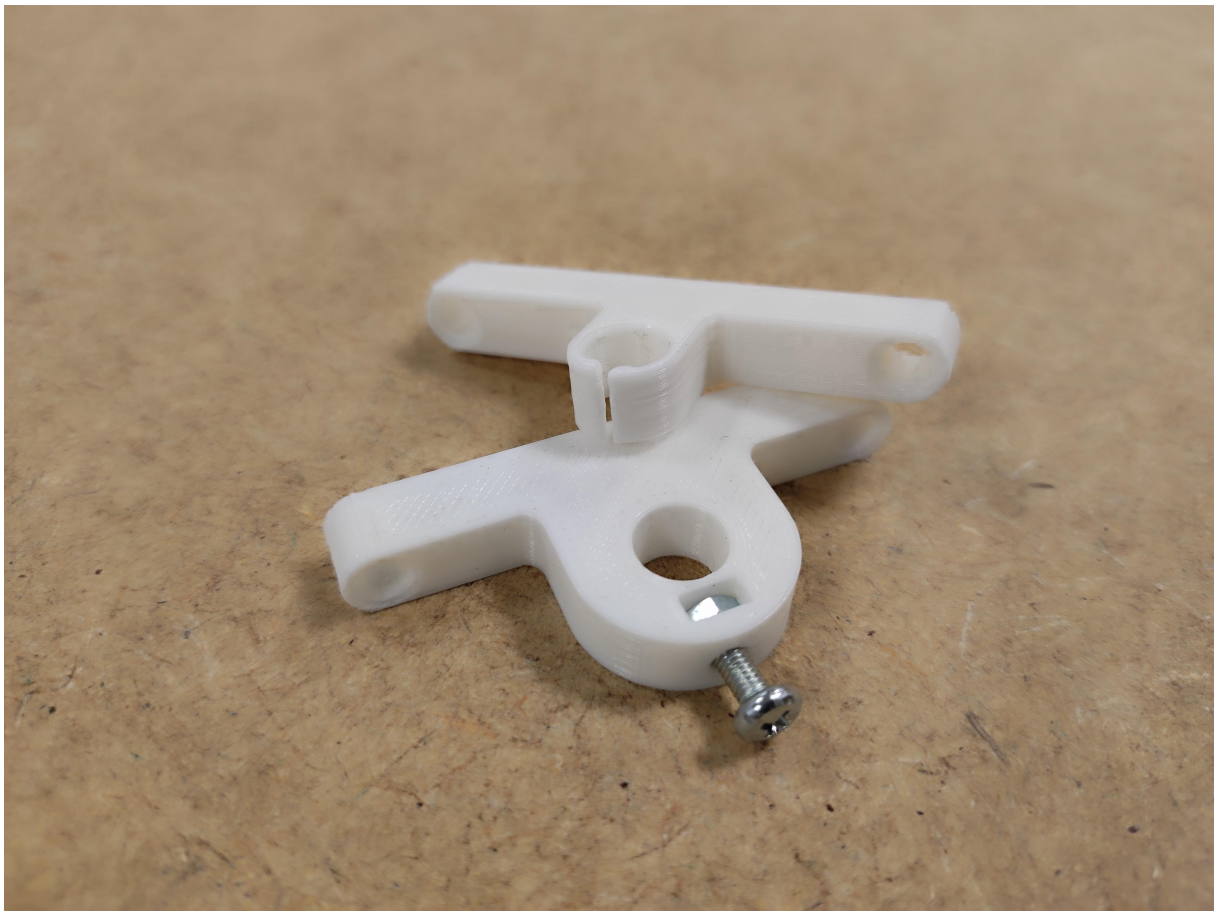
Otto's Pen Adapter is an easy-to-print accessory that gives your HP Robots Otto the ability to make drawings using a pen. Different pen sizes adapters are available, from BIC ballpoint pens to Posca pens (see STL files for pen reference).

There are two adapters, a spring-loaded grip mechanism and a screw-locking mechanism that secures the pen in place. Both designs serve the same purpose, ensuring the pen doesn't wobble during use, but they handle mechanical tension differently and with different parts number.



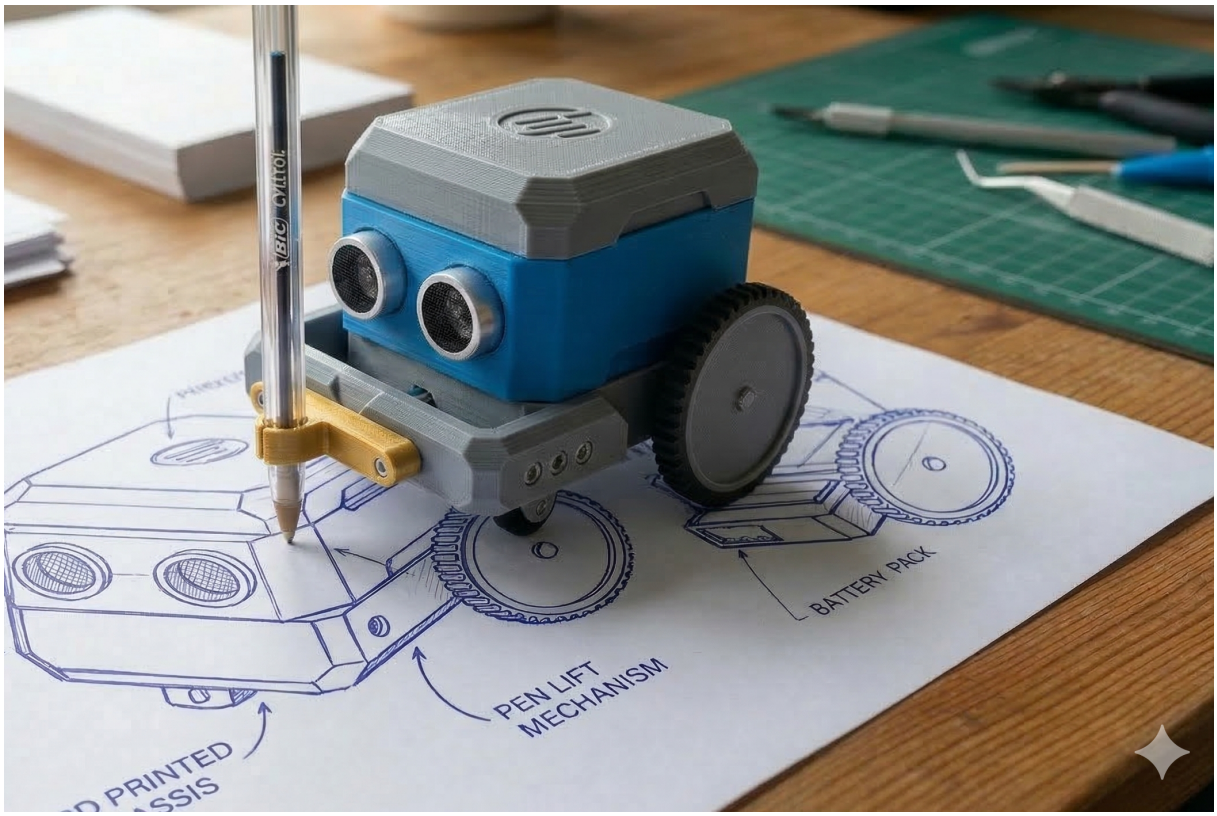
The pen adapter mounts on the bumper of the **Otto Gripper** and it's fully compatible with the Otto modular system.

Print it with PLA or any other rigid filament, with no supports, layer height <=0.2mm and infill >=15%.



The pen adapter was modeled in OpenSCAD, and I've 3D printed the two versions for a BIC ballpoint pen. The OpenSCAD file is available. The design is parametric so that you can tailor the pen adapter for your needs and for any given diameter pen. The OpenSCAD script needs the 'hp_gripperbase_r1.stl' file from the 'Otto Gripper' which I've included here for completeness.

Because I don't have a HP Otto to test it, I've added a Gemini AI rendering of the final version for a BIC ballpoint pen.



Let's draw!

Personal note on the contest results

It was fun making this entry for the contest, but I think that the competition was not totally fair, the winning models are of little to no use to the target audience that uses Otto: children, teenagers, and people who actually want to learn robotics with 3D printing. In the end it looks like an exploitative marketing gimmick using the competitors' work.

Model files



pc-17k.stl



pc-3m.stl



pc-5m.stl



pc-8k.stl



pc-1m.stl



bic-cristal.stl



pc-1mr.stl



hp_gripperbase_r1.stl



ottopenadapterwithscrew.stl



ottopenadapter.scad

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